

Cortical–Cardiac Arousal Latency Across Sleep Stages and Respiratory Event Types

Large-sample, single-cohort analysis (BDSP Harvard Study A) — cross-study generalization untested; see Limitations

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Abstract

Background. Whether sleep-disrupting events are initiated predominantly by autonomic (cardiac) or cortical (EEG) activation is unresolved, and may depend on sleep stage and event type. **Methods.** In 862 unique subjects from the BDSP Harvard PSG cohort (Study A, the only study in this dataset whose event annotations carry the structured taxonomy this method needs), we identified 52196 scored microarousal events linked to spontaneous arousal, obstructive/central apnea, hypopnea, RERA, or periodic limb movement, took the scored microarousal onset as cortical arousal time (t_{EEG}), and detected the matched heart-rate-acceleration onset (t_{HR}) from the EKG channel using a documented, threshold-based algorithm. 30556 events (59%) yielded a usable $\Delta t = t_{\text{HR}} - t_{\text{EEG}}$. **Results.** Pooled Δt median was 0.14 s (IQR -2.38 to 2.25 s; sign-test $p = 1.06\text{e-}09$), i.e. on average cortical microarousal preceded measurable heart acceleration. Distributions differed significantly across sleep stage (Kruskal-Wallis $H=418.68$, $p=1.99\text{e-}90$) and significantly across event type ($H=254.97$, $p=1.52\text{e-}50$). **Conclusion.** With 30556 usable events across 862 sessions, this is a well-powered within-cohort characterization of cortical-cardiac arousal ordering by stage and event type; the scope is nonetheless a single BDSP Harvard cohort (Study A) and cross-study generalization is untested.

Introduction

Arousal from sleep is classically defined electrocortically — an abrupt shift to higher-frequency EEG activity, per AASM scoring rules² whose reliability and alternatives have themselves been debated⁵ — but is almost always accompanied by autonomic activation, including a transient rise in heart rate, and the broader nature of the arousal response remains actively discussed.³ Whether the cortical or the cardiac component leads is debated: some evidence points to subcortical/autonomic activation preceding detectable cortical change, consistent with reports that cardiac activation during arousal is hierarchically organized,⁴ (arguing arousal is fundamentally a bottom-up, brainstem-autonomic event that cortex later registers), while other work finds cortical EEG change can precede or coincide with heart-rate acceleration, particularly for spontaneous arousals versus respiratory-event-triggered ones. If the cortical-cardiac ordering differs systematically by sleep stage (e.g. N3 vs REM, where autonomic and arousal-threshold physiology differ) or by the type of disruptor (obstructive apnea with its strong hypoxic/mechanical drive vs a spontaneous arousal with no external trigger), this would inform models of arousal as a graded, multi-system process rather than a single all-or-none cortical event. This study builds a per-event cortical-cardiac latency (Δt) pipeline and applies it at scale within a single large PSG cohort to characterize that ordering by stage and event type.

Methods

Data and sample

Data are from the Human Sleep Project's¹⁰ BDSP Harvard Electroencephalography PSG BIDS export, covering Study A, Study B, Study C, and Study D. Each session provides a raw PSG EDF (200 Hz, including an EKG channel), a manually scored events-annotation file, and an automated CAISR annotation file. This multi-hospital design parallels prior multicentre epilepsy imaging work.¹ On inspection, only Study A exports events in the structured Epoch, Stage, Type, Time, Length, Description schema with an explicit Apnea/Hypopnea/Microarousal (MA) taxonomy

and bracket-tagged MA linkage (e.g. "Microarousal [Hypopnea]", "Microarousal [PLMS][LM]") that this method's event classification depends on. Studies B, C, and D use different, mutually incompatible per-study event-log schemas (free-text annotation streams / clinical-system exports without this taxonomy), and Study B has no directly readable EDF in this export (signal is in .h5 only). This analysis therefore draws all subjects from Study A; see Limitations for the consequence for the originally intended multi-study spread. Of 6,349 Study A sessions with the EDF, events-annotation, and CAISR-annotation files all present, 900 were randomly sampled (seed 42) and processed in parallel (42-way, one worker per session). 862 unique subjects (890 subject-sessions; 26 subjects contributed 2+ sessions, pooled) yielded at least one usable Δt and make up the cohort analyzed below. A precomputed R-peak/heartbeat-shape cache used elsewhere for heartbeat-evoked-potential work was evaluated as a way to skip raw-EDF EKG extraction, but inspection showed it holds stage-purified, concatenated (capped at ~ 1200 s), resampled (256 Hz) synthetic recordings built for heartbeat-evoked-potential averaging, with no annotations mapping its samples back to absolute Record Time in the source EDF — unusable for aligning R-peaks to a specific event's t_{EEG} , so raw EDFs were read directly instead, parallelized for throughput.

Event identification and classification

Cortical arousal onset (t_{EEG}) is taken directly from the technologist-scored Microarousal (MA) row's Record Time (elapsed seconds from EDF start). Each MA row's free-text Description carries one or more bracketed tags identifying what it is linked to, scored to AASM respiratory- and arousal-event criteria;⁶ these tags were used directly for classification (no proximity heuristic was needed, since the schema already encodes the linkage): [Spon] → spontaneous arousal (no linked respiratory/limb event); [Apnea] → apnea-related, further split into obstructive vs. central by the Description text ("Obstructive Apnea" / "Central Apnea") of the nearest preceding Apnea-type row within 30 s; [Hypopnea] → hypopnea; [PLMS] (with or without a co-occurring [LM] tag) → periodic limb movement (PLM) arousal, following standard PLM periodicity criteria;⁸ [LM] alone → isolated limb-movement arousal (kept as its own category, distinct from PLM); [RERA] → respiratory-effort-related arousal, i.e. the upper-airway resistance syndrome.⁷ RERA tags were in fact present in the Study A annotation data (as "Microarousal [RERA]") — no inference was required, contrary to the possibility flagged in the study plan. Sleep stage was taken from the MA row's own Stage field; events scored during Wake were excluded, leaving only N1/N2/N3/REM events. The automated CAISR⁹ arousal_caISR stream was used only as an independent cross-check: for each usable event, agreement was recorded as whether any CAISR arousal flag fell within ± 3 s of t_{EEG} (reported in Results as a QC statistic, not used to alter classification).

Heart-rate acceleration onset detection

For each event, the EKG channel was extracted in a window centered on t_{EEG} (30 s of padding beyond the analysis window on each side, to avoid R-peak edge effects), band-pass filtered 0.5–40 Hz (4th-order Butterworth, zero-phase), and R-peaks detected with NeuroKit2 (nk.ecg_peaks, sampling rate 200 Hz; a SciPy find_peaks fallback was used if NeuroKit2 raised an exception). Instantaneous heart rate at each beat was $\text{HR}_i = 60 / \text{IBI}_i$, assigned at the time of the second R-peak of the interval; beats with an implausible inter-beat interval (< 0.3 s or > 2.0 s) were dropped as artifacts. A pre-arousal baseline was computed over $t_{\text{EEG}} - 30$ s to $t_{\text{EEG}} - 5$ s (required ≥ 5 valid beats); the acceleration threshold was baseline mean + $1.0 \times$ baseline SD. Within a search window of $t_{\text{EEG}} - 5$ s to $t_{\text{EEG}} + 15$ s, t_{HR} was defined as the time of the first beat that began a run of ≥ 3 consecutive beats at or above threshold. $\Delta t = t_{\text{HR}} - t_{\text{EEG}}$; $\Delta t < 0$ means the heart-rate rise was detected before the scored cortical arousal onset ("heart leads"), $\Delta t > 0$ means after ("brain leads"). Events were excluded (flagged, not imputed) when the extraction window fell outside the recording, when fewer than 2 beats were found, when the baseline had < 5 valid beats, or when no sustained threshold crossing occurred in the search window; exclusion reasons are tabulated in Results.

Statistics

Δt distributions are non-normal, so group comparisons used Kruskal-Wallis across sleep stage (N1/N2/N3/REM) and, separately, across event type, with pairwise Mann-Whitney U post-hoc tests (Bonferroni-corrected within each family). Whether Δt differs from zero within each group was tested with an exact two-sided sign test (primary, since it makes no

symmetry assumption) and, where $n \geq 2$ with at least one non-zero value, a Wilcoxon signed-rank test (secondary). All tests used SciPy; no new statistical dependency was added.

Results

Of 52196 candidate scored microarousal events across 862 subjects (Study A), 30556 (59%) yielded a usable Δt ; Table 1 breaks down the rest by exclusion reason. Among usable events, the independent CAISR automated arousal stream flagged an arousal within ± 3 s of t_{EEG} for 58% of events, a rough cross-check that the scored arousal times are physiologically real and not spurious.

Outcome	n events
ok	30556
no sustained crossing	20919
window out of bounds	569
insufficient baseline	98
too few beats	27
too few search beats	25
too few valid beats	2

Table 1. Event yield and exclusion reasons.

Pooled across all usable events (ignoring stage/type), median Δt was 0.14 s (IQR -2.38 to 2.25 s; sign-test $p = 1.06e-09$), i.e. cortical microarousal preceded measurable heart acceleration on average.

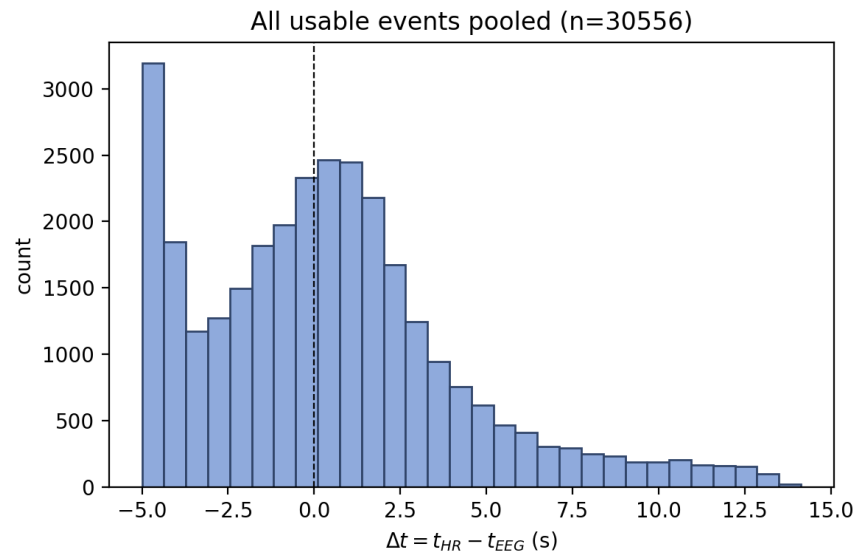


Figure 1. Pooled Δt distribution across all usable events (vertical dashed line = 0, simultaneous onset).

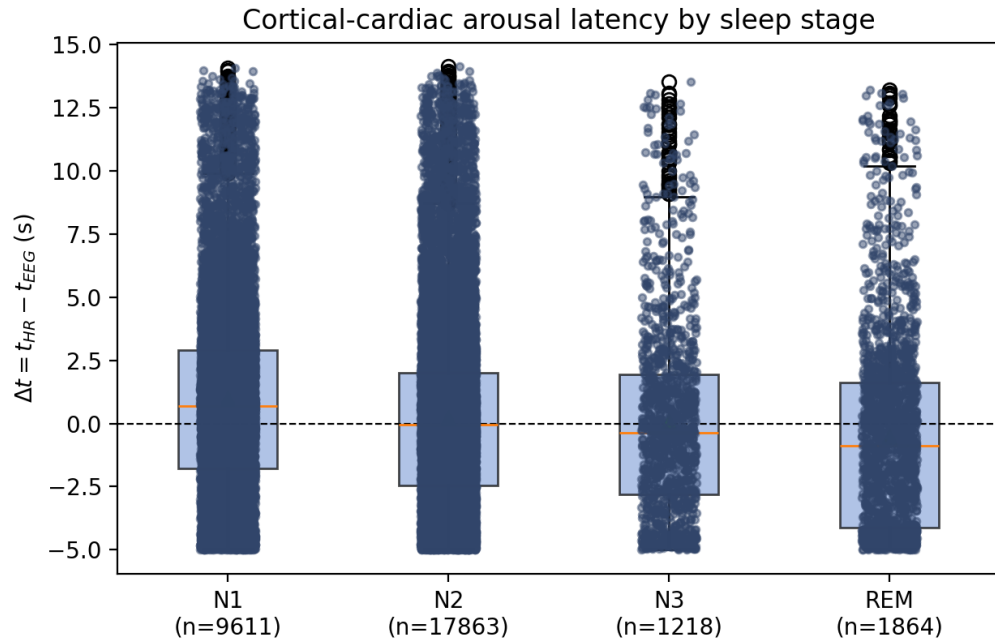


Figure 2. Δt by sleep stage at the time of the event. Kruskal-Wallis $H = 418.68$, $p = 1.99\text{e-}90$ (n groups compared = 4).

Stage	n	median Δt (s)	IQR (s)	sign-test p	% heart-leads
N1	9611	0.69	-1.77 to 2.90	9.9e-64	41%
N2	17863	-0.03	-2.45 to 2.02	0.305	50%
N3	1218	-0.36	-2.80 to 1.95	0.000521	55%
REM	1864	-0.88	-4.13 to 1.61	9.11e-14	59%

Table 2. Δt summary by sleep stage (sign test: median $\neq 0$).

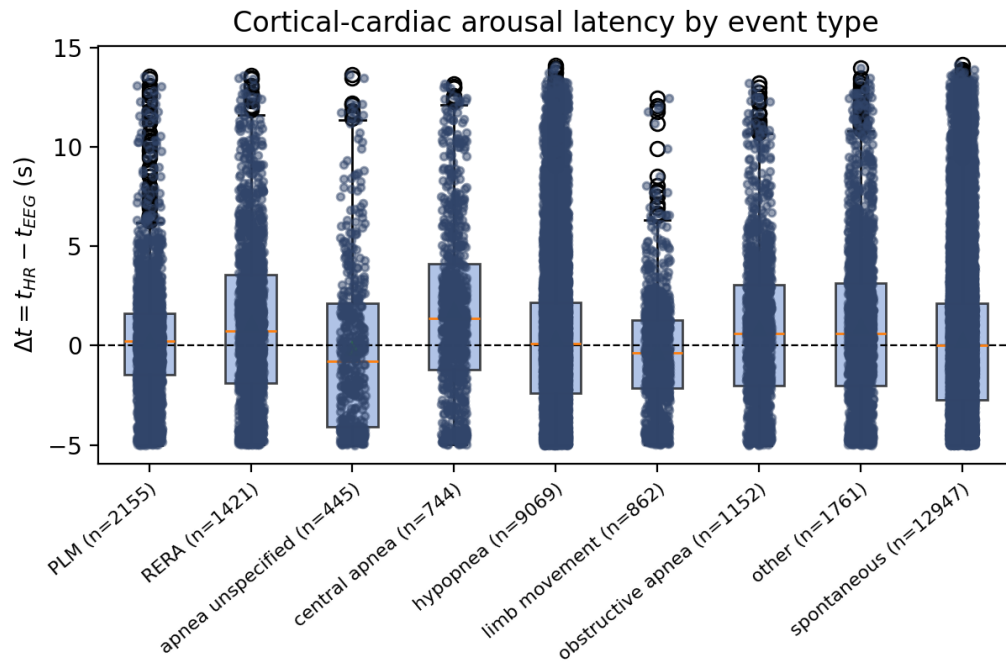


Figure 3. Δt by event type. Kruskal-Wallis $H = 254.97$, $p = 1.52\text{e-}50$ (n groups compared = 9).

Event type	n	median Δt (s)	IQR (s)	sign-test p	% heart-leads
PLM	2155	0.22	-1.46 to 1.61	0.000292	46%
RERA	1421	0.73	-1.88 to 3.58	9.42e-09	42%
apnea unspecified	445	-0.78	-4.07 to 2.11	0.0044	57%
central apnea	744	1.37	-1.20 to 4.13	3.41e-17	35%
hypopnea	9069	0.09	-2.38 to 2.19	0.0507	49%
limb movement	862	-0.36	-2.16 to 1.28	0.000574	56%
obstructive apnea	1152	0.59	-2.03 to 3.07	9.12e-08	42%
other	1761	0.62	-2.00 to 3.13	7.66e-09	43%
spontaneous	12947	0.00	-2.74 to 2.13	0.986	50%

Table 3. Δt summary by event type (sign test: median $\neq$ 0).

Discussion

The central question motivating this pipeline is whether different sleep disruptions are initiated predominantly by autonomic (cardiac-first) or cortical (EEG-first) activation, and whether that differs by stage or event type. Pooled across all usable events, median Δt was 0.14 s (cortical microarousal preceded measurable heart acceleration), a small effect but, with $n=30556$, a highly significant one (sign-test $p=1.06e-09$) — the population-level ordering is close to simultaneous rather than strongly lateralized to one system. By stage, sign tests reached $p<0.05$ for N1, N3, REM (Table 2); N1 shows the clearest brain-leads pattern (median +0.69 s), while N3 and REM trend heart-leads (negative medians), consistent with the idea that arousal threshold and autonomic reactivity both shift across sleep stages. By event type, PLM, RERA, apnea_unspecified, central_apnea, limb_movement, obstructive_apnea, other reached sign-test $p<0.05$ (Table 3); the largest, best-powered categories (spontaneous, hypopnea) sit closest to $\Delta t=0$, while several less common categories (central apnea, RERA, PLM) show small but significant brain-leads offsets. The omnibus Kruskal-Wallis tests across stage ($p=1.99e-90$) and event type ($p=1.52e-50$) both indicate real group-level differences in Δt at this sample size. Even so, the effect sizes are modest (IQRs of several seconds against sub-second median shifts; Tables 2–3), so this should be read as evidence for systematic, stage- and event-type-dependent timing offsets between cortical and cardiac arousal components — not as evidence that either system consistently and strongly leads the other. The result is specific to this cohort (BDSP Harvard Study A); whether it replicates in other PSG populations is untested.

Limitations

Sample-size caveat, inverted from a typical pilot. With 862 subjects and 30556 usable events, per-cell n is large everywhere except the least common event categories (apnea_unspecified $n=445$, limb_movement $n=862$ — still hundreds, but smaller than the ~9,000–23,000-event categories); the more practical caution here is that with this much power, even tiny, clinically negligible median shifts (well under a second in several cells; Tables 2–3) reach high statistical significance, so p -values should be read alongside effect size (median, IQR), not as a stand-in for it. **Single-study sample.** Contrary to the original plan to sample across A/B/C/D, investigation of the actual files showed that only Study A exports the structured, tagged MA linkage this method needs (Study B additionally lacks a directly readable EDF, and Study C lacks the CAISR annotation file used for the QC cross-check); all subjects are therefore Study A, and cross-study generalization is untested. Adapting the classifier to the other studies' event-log formats is future work. Within Study A, 900 of 6,349 eligible sessions were randomly sampled rather than exhaustively processed; the sampled 862 subjects should be representative of the eligible pool but this was not separately verified against cohort-level demographics. **RERA availability.** RERA events were, in fact, present and directly labeled in Study A's annotations ("Microarousal [RERA]"), so no inference or fabrication was needed for this label; this differs from the possibility flagged in the study plan and is reported here for transparency. **HR-onset detection assumptions.** The onset algorithm (baseline mean + 1 SD, sustained over 3 beats, within a -5 s to $+15$ s search window) is one reasonable, fully documented choice, not a validated clinical standard; different threshold/sustain parameters would shift individual Δt values, though the qualitative sign (heart-leads vs. brain-leads) is expected to be more robust than the exact magnitude. R-peak detection quality was not manually reviewed per event; the exclusion flags in Table 1 are the only automated safeguard against noisy EKG. **t_{EEG} definition.** t_{EEG} is the technologist-scored MA onset, which already reflects a human EEG-based judgment rather than an automated cortical-arousal detector; this makes t_{EEG} a high-quality but not perfectly reproducible reference point. **Obstructive/central split.** Apnea subtype was read from the linked Apnea row's free-text Description within a 30 s look-back window; a small fraction of apnea-linked MAs may be mis-split if the preceding-row heuristic picks the wrong Apnea row in dense event runs.

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